

Omni-Cloud Security Project

AWS + Azure Implementation

Comprehensive Multi-Cloud Security Documentation

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Total Screenshots: 75 (65 AWS + 10 Azure)

Total Phases: 5

Project Cost: \$2.60 (with cleanup)

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All screenshots are available at : [COMPLETE SCREENSHOT GALLERY](#)

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1 Project Overview

1.1 Executive Summary

This document provides comprehensive documentation for the Omni-Cloud Security Project, a multi-cloud security implementation across AWS and Azure. The project demonstrates infrastructure deployment, security controls, DevSecOps pipeline implementation, attack simulation, SOAR automation, and compliance monitoring. GCP has been excluded from this implementation.

1.2 Project Objectives

- Deploy secure infrastructure on AWS and Azure
- Implement comprehensive security controls including IAM, WAF, and Secrets Management
- Build a DevSecOps pipeline with automated security scanning
- Simulate real-world attacks and implement automated remediation (SOAR)
- Establish compliance monitoring using AWS Config
- Optimize costs through resource management

1.3 Project Scope

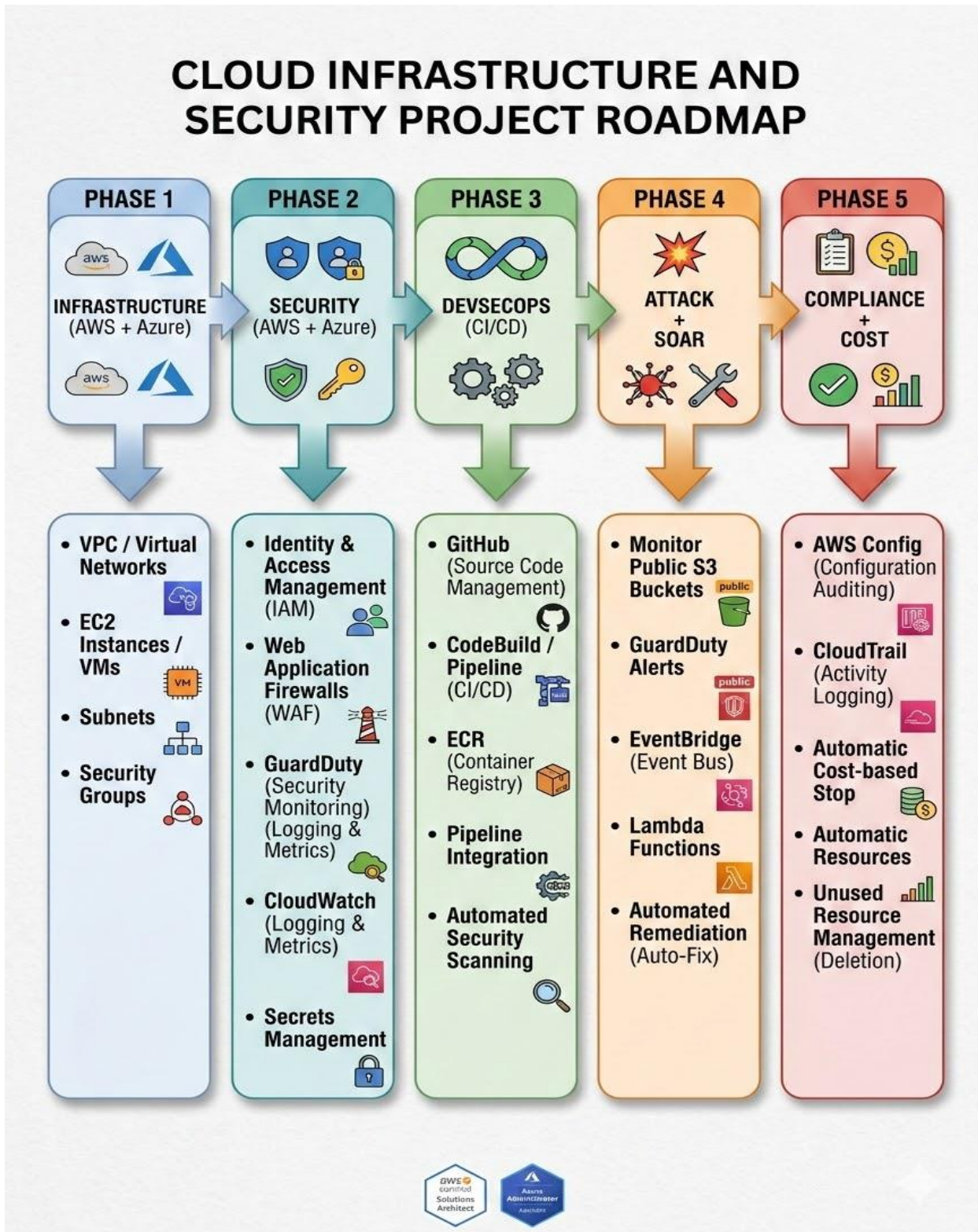
- **AWS Services Used:** VPC, EC2, IAM, WAF, ALB, Secrets Manager, Guard Duty, CloudWatch, SNS, Lambda, Event Bridge, CloudTrail, S3, Code Build, CodePipeline, ECR, AWS Config (20+ services)
- **Azure Services Used:** Resource Group, Virtual Network, Network Security Group, Virtual Machines, Activity Log, Metrics, Alerts (5+ services)
- **Total Screenshots:** 75 (65 AWS + 10 Azure)

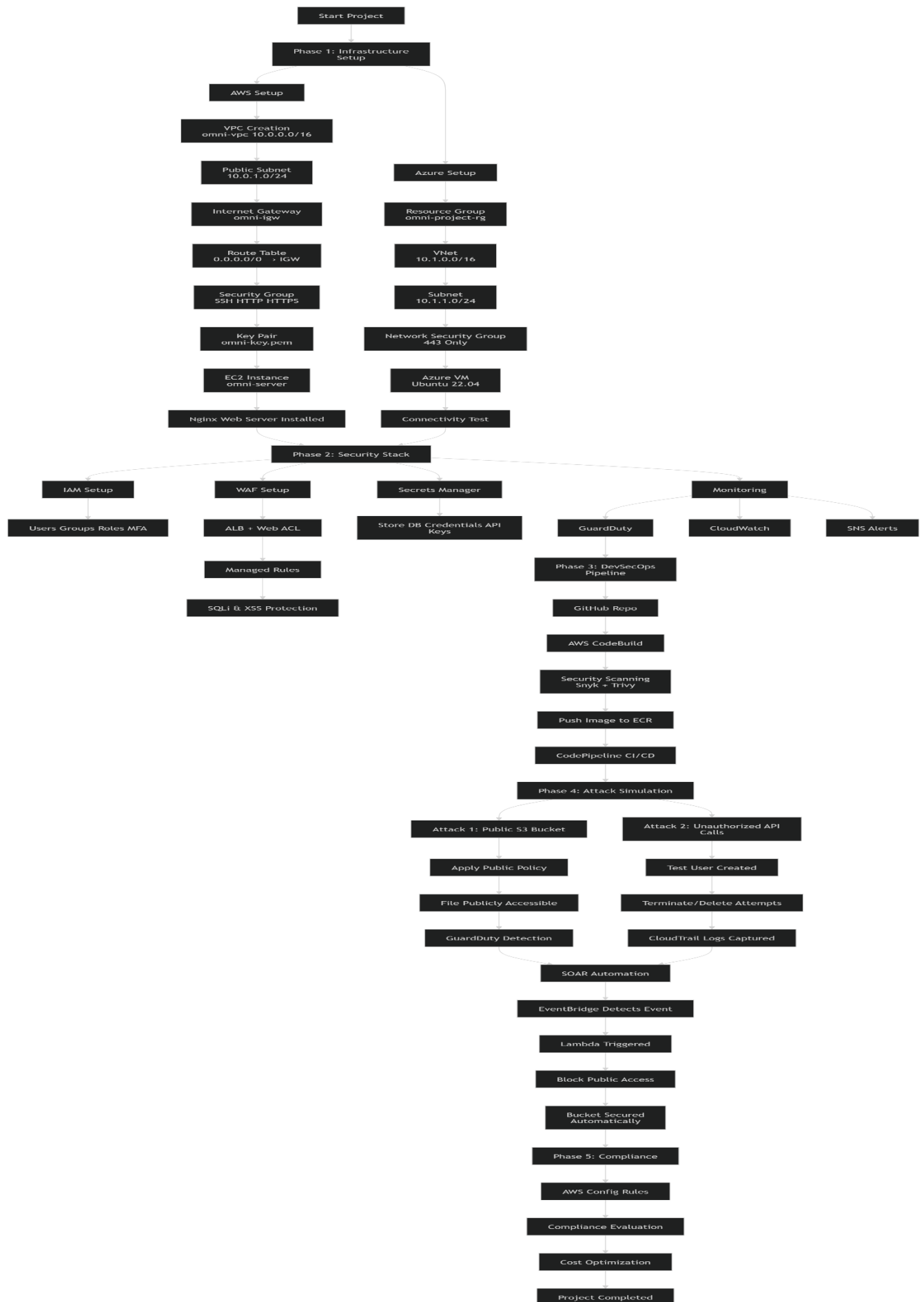
1.4 Project Timeline

- **Phase 1:** Foundation - Infrastructure Setup
- **Phase 2:** Security Stack - IAM, WAF, Secrets, Monitoring
- **Phase 3:** DevSecOps - CI/CD Pipeline & Container Security
- **Phase 4:** Attack Simulation & SOAR Automation
- **Phase 5:** Compliance | Cost Optimization

2 Complete Project Workflow Diagram

2.1 Visual Project Architecture





2.2 Diagram Components Summary

- **Phase 1:** AWS VPC/EC2 Setup + Azure VNet/VM Setup
- **Phase 2:** IAM, WAF, Secrets Manager, Guard Duty, CloudWatch + Azure Monitoring
- **Phase 3:** GitHub, Code Build, ECR, CodePipeline CI/CD Pipeline
- **Phase 4:** Attack Simulation (Public S3 + Unauthorized API) and SOAR Automation
- **Phase 5:** AWS Config Compliance and Cost Optimization

3 Phase 1: Foundation - Infrastructure Setup

3.1 Overview

Phase 1 establishes the foundational infrastructure on both AWS and Azure platforms. This phase ensures proper network isolation, security group configuration, and compute resource deployment.

3.2 AWS Infrastructure Setup

3.2.1 Virtual Private Cloud (VPC) Configuration

- **VPC Name:** omni-vpc
- **CIDR Block:** 10.0.0.0/16
- **DNS Resolution:** Enabled
- **DNS Hostnames:** Enabled
- **Region:** US East (N. Virginia) us-east-1

Screenshot Reference: AWS-01-VPC-Dashboard.png - VPC dashboard showing omni-vpc

3.2.2 Public Subnet Configuration

- **Subnet Name:** omni-public-subnet
- **CIDR Block:** 10.0.1.0/20
- **Availability Zone:** us-east-1a
- **Auto-assign Public IP:** Yes

Screenshot Reference: AWS-02-Subnet.png - Public subnet configuration.

3.2.3 Internet Gateway Setup

- **Internet Gateway Name:** omni-igw
- **Attachment:** Attached to omni-vpc

Screenshot Reference: Implicit in route table configuration.

3.2.4 Route Table Configuration

- **Route Table Name:** omni-public-rtb-public
- **Routes:**
 - 10.0.0.0/16 → local
 - 0.0.0.0/0 → Internet Gateway (igw- o6b9d4e8a8655df2f)
- **Association:** Associated with omni-public-subnet

Screenshot Reference: AWS-03-Route-Table.png - Route table with IGW route

3.2.5 Security Group Configuration

- **Security Group Name:** omni-sg
- **Inbound Rules:**
 - SSH (TCP 22) from custom IP
 - HTTP (TCP 80) from 0.0.0.0/0
 - HTTPS (TCP 443) from 0.0.0.0/0
- **Outbound Rules:** All traffic to 0.0.0.0/0

Screenshot Reference: AWS-04-Security-Group.png - Security group inbound rules

3.2.6 Key Pair Creation

- **Key Pair Name:** omni-key
- **Type:** RSA
- **Format:** .pem
- **Fingerprint:** b1:35:ce:68:18:8c:09:e5:7c:a8:87:50:0...

Screenshot Reference: AWS-05-Key-Pair.png - Key pair omni-key created.

3.2.7 EC2 Instance Launch

- **Instance Name:** omni-server
- **Instance ID:** i-oa04784172f813512.
- **Instance Type:** t2.micro (Free tier eligible)
- **AMI:** Amazon Linux 2023
- **Public IP:** 98.84.36.103
- **Private IP:** 10.0.3.74
- **State:** Initially RUNNING, later STOPPED
- **Root Volume:** 20 GB gp3

Screenshot References:

- AWS-06-EC2-Running.png - EC2 instance in RUNNING state
- AWS-09-EC2-Stopped.png - EC2 instance in STOPPED state

3.2.8 Web Server Installation

```
1 sudo dnf update -y
2 sudo dnf install nginx -y
3 sudo systemctl start nginx
4 sudo systemctl enable nginx
5 sudo systemctl status nginx
```

Listing 1: nginx Installation Commands

Screenshot References:

- AWS-07-EC2-SSH-Nginx-Install.png - SSH session with nginx installation
- AWS-08-Nginx-Welcome-Page.png - Browser showing nginx welcome page

3.3 Azure Infrastructure Setup

3.3.1 Resource Group Creation

- **Resource Group Name:** omni-project-rg
- **Region:** East US 2
- **Subscription:** Azure for Students
- **Resources:** 6 total (NSG, VM, Public IP, NIC, Disk, VNet)

Screenshot Reference: Azure-01-Resource-Group.png - Resource group with all resources

3.3.2 Virtual Network (VNet) Configuration

- **VNet Name:** omni-vnet
- **Address Space:** 10.1.0.0/16
- **Region:** East US 2
- **DNS Servers:** Azure-provided

Screenshot Reference: Azure-02-VNet.png - Virtual network omni-vnet

3.3.3 Subnet Configuration

- **Subnet Name:** omni-subnet
- **Address Prefix:** 10.1.1.0/24
- **Network Security Group:** omni-nsg

3.3.4 Network Security Group (NSG) Configuration

- **NSG Name:** omni-nsg
- **Inbound Rules:**
 - **Priority 100:** SECURE-WEB-ACCESS - Port 443, TCP, Any → Any, Allow
 - **Priority 65000:** AllowVnetInBound (default)
 - **Priority 65001:** AllowAzureLoadBalancerInBound (default)
 - **Priority 65500:** DenyAllInBound (default)
- **Design Philosophy:** ONLY port 443 allowed (principle of least privilege)

Screenshot Reference: Azure-03-NSG-Rules.png - NSG inbound rules showing ONLY port 443

3.3.5 Azure Virtual Machine Deployment

- **VM Name:** omni-vm
- **Size:** Standard B1s (1 vCPU, 1 GB RAM)
- **OS:** Ubuntu 22.04 LTS (Gen2)
- **Username:** admin5911
- **Authentication:** Password (due to IAM limitations)
- **Public IP:** 74.249.54.234
- **State:** Initially RUNNING, later STOPPED (deallocated)
- **Availability Zone:** 2

Screenshot References:

- Azure-04-VM-Running.png - VM in RUNNING state
- Azure-06-VM-Stopped.png - VM in STOPPED (deallocated) state

3.3.6 Connectivity Testing

```

1 nc -zv 74.249.54.234 443 # HTTPS - Should be OPEN
2 nc -zv 74.249.54.234 22  # SSH - Should be BLOCKED
3 nc -zv 74.249.54.234 80  # HTTP - Should be BLOCKED

```

Listing 2: Port Connectivity Tests

Test Results:

- **Port 443:** ✓OPEN - Connection succeeded.
- **Port 22:** ✗BLOCKED - Connection timed out.
- **Port 80:** ✗BLOCKED - Connection timed out.

Screenshot Reference: Azure-05-Port-443-Test.png - Port 443 connection successful.

4 Phase 2: Security Stack Implementation

4.1 Overview

Phase 2 implements comprehensive security controls across both clouds, including Identity and Access Management (IAM), Web Application Firewall (WAF), Secrets Management, and Monitoring.

4.2 AWS Identity and Access Management (IAM)

4.2.1 IAM Groups

- **Group Name:** omni-admin
- **Attached Policies:** Administrator Access
- **Purpose:** Administrative access for project management

Screenshot Reference: AWS-10-IAM-Group.png - User group omni-admins with AdministratorAccess

4.2.2 IAM Users

Username	Permissions	Purpose
omni-operator	Administrator Access	Full administrative access
omni-viewer	omni-read-only policy	Read-only access for monitoring

Table 1: IAM Users Created

Screenshot References:

- AWS-11-IAM-User-Operator.png - User omni-operator created.
- AWS-12-IAM-Operator-Credentials.csv - Credentials for omni-operator
- AWS-13-IAM-Viewer-Credentials.csv - Credentials for omni-viewers
- AWS-14-IAM-Users-List.png - Both IAM users listed.

4.2.3 Custom IAM Policies

Policy Name: omni-readonly

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "ec2:Describe*",
8         "s3:List*",
9         "s3:Get*",

```

```

10         "cloudwatch:Describe *",
11         "cloudwatch:Get *",
12         "cloudwatch:List *"
13     ],
14     "Resource": "*"
15 }
16 ]
17 }

```

Listing 3: omni-readonly Policy JSON

Screenshot References:

- AWS-15-IAM-JSON-Policy.png - Custom JSON policy for omni-readonly
- AWS-16-IAM-Omni-Readonly-Policy.png - omni-readonly policy created

4.2.4 IAM Roles

- **Role Name:** omni-ec2-s3-readonly
- **Purpose:** Allows EC2 instances to read S3 buckets
- **Trust Entity:** EC2 service
- **Attached Policy:** AmazonS3ReadOnlyAccess

Screenshot Reference: AWS-17-IAM-Role.png - Role omni-ec2-s3-readonly created

4.2.5 Multi-Factor Authentication (MFA)

- **MFA Device:** Virtual (Authapp)
- **ARN:** arn:aws:iam::778754217204:mfa/Authapp
- **Enabled On:** Root account

Screenshot Reference: AWS-36-MFA-Enabled.png - MFA enabled on root account

4.2.6 IAM Least Privilege Testing

Test Performed: Attempt to create EC2 instance with omni-viewer credentials

Result: ✗Access Denied - "User is not authorized to perform ec2:CreateSecurityGroup"

Screenshot Reference: AWS-37-IAM-Test-Denied.png - omni-viewer denied EC2 creation

4.3 AWS Web Application Firewall (WAF)**4.3.1 Application Load Balancer (ALB) Setup**

- **ALB Name:** omni-alb
- **Scheme:** Internet-facing
- **DNS Name:** omni-alb-745421209.us-east-1.elb.amazonaws.com

- **VPC:** omni-vpc
- **Security Groups:** sg-0ce510f82aab86b2e
- **Target Group:** omni-tg (EC2 instance)

Screenshot References:

- AWS-18-ALB-Created.png - Application Load Balancer omni-alb
- AWS-19-ALB-Test.png - ALB showing nginx welcome page

4.3.2 Web ACL Configuration

- **Web ACL Name:** omni-web-acl
- **Total Rules:** 19
- **Scope:** Regional (ALB)
- **Default Action:** Allow

AWS Managed Rules Enabled:

- Core rule set (CRS) - OWASP Top 10 protection
- SQL database - SQL injection protection
- Known bad inputs - Blocks malicious patterns
- Amazon IP reputation list - Blocks known malicious IPs
- AWS-AWSManagedRulesAntiDDoSRuleSet
- AWS-AWSManagedRulesBotControlRuleSet
- AWS-AWSManagedRulesLinuxRuleSet

Custom Rules:

- Rate-based rule (2000 requests/5 minutes)
- GeoRule
- BodySizeRestrictionRule

Screenshot Reference: AWS-20-WAF-Rules.png - Web ACL with 19 rules

4.3.3 WAF Testing - SQL Injection

```
1 curl "http://omni-alb-745421209.us-east-1.elb.amazonaws.com/?id=1%27%20R%20%271%27=%271"
```

Listing 4: SQL Injection Test

Response:

```
1 <html >
2 <head ><title >403 Forbidden </title ></head >
3 <body >
4 <center ><h1 >403 Forbidden </h1 ></center >
5 </body >
6 </html >
```

Screenshot Reference: AWS-21-WAF-Test-403.png - 403 Forbidden from SQL injection.

4.4 AWS Secrets Manager

4.4.1 Secret Creation

- **Secret Name:** omni/app/secrets
- **Encryption Key:** aws/secretsmanager
- **Secret ARN:** arn:aws:secretsmanager:us-east-1:778754217204:secret:omni/app/secrets-g9YNRR

Secret Values Stored:

Key	Value
db username	omni-admin
db password	SuperSecure123!
api key	sk-live-abc123xyz789
connection string	postgresql://omni_admin:SuperSecure123!@database.omni.com:5432

Table 2: Secrets Stored in AWS Secrets Manager

Screenshot References:

- AWS-31-Secrets-Manager.png - Secret omni/app/secrets stored.
- AWS-32-Secrets-Values.png - Secret values displayed.

4.4.2 Secrets Retrieval Testing

CLI Command:

```
1 aws secretsmanager get-secret-value --secret-id omni/app/secrets --query. '
  SecretString' --output text
```

Listing 5: Secret Retrieval via AWS CLI

Output:

```

1 {
2   "db_username": "omni_admin",
3   "db_password": "SuperSecure123!",
4   "api_key": "sk-live-abc123xyz789",
5   "connection_string": "postgresql://omni_admin:SuperSecure123!@database.
   omni.com:5432/prod"
6 }

```

Screenshot Reference: AWS-33-Secrets-CLI.png - CLI retrieval of secrets

4.4.3 Secrets Access Policy

Policy Name: omni-secret-readonly

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "secretsmanager:GetSecretValue",
8         "secretsmanager:DescribeSecret"
9       ],
10      "Resource": "arn:aws:secretsmanager:us-east-1:
      :778754217204:secret:omni / app / secrets -*"
11    }
12  ]
13 }

```

Listing 6: Secrets Read-Only Policy

Screenshot References:

- AWS-34-Secrets-Policy.png - omni-secret-readonly policy
- AWS-35-Secrets-Policy-Attached.png - Policy attached to user

4.5 AWS Guard Duty Threat Detection

4.5.1 Guard Duty Enablement

- **Status:** Enabled
- **Detector ID:** 04ce5c0e3860b34a7d907d368ced9222
- **Free Trial:** 30 days for EC2, EKS, Fargate
- **Findings Export:** Every 15 minutes to Event Bridge

Screenshot References:

- AWS-22-GuardDuty-Dashboard.png - Guard Duty dashboard
- AWS-24-GuardDuty-Settings.png - Guard Duty settings

4.5.2 Runtime Monitoring

- **Status:** Enabled
- **Coverage:** EC2 instances
- **Free Trial:** 30 days

Screenshot Reference: AWS-23-GuardDuty-Runtime.png - Runtime Monitoring enabled

4.6 AWS CloudWatch Monitoring

4.6.1 SNS Notifications Setup

- **Topic Name:** omni-alerts
- **Display Name:** OMNI ALERTS
- **Protocol:** Email
- **Endpoint:** sr9063287@gmail.com
- **Status:** Confirmed

Screenshot References:

- AWS-25-SNS-Topic.png - SNS topic omni-alerts created
- AWS-26-SNS-Email-Confirmation.png - Email confirmation request
- AWS-27-SNS-Subscription.png - Subscription confirmed

4.6.2 CloudWatch Alarms

Alarm 1: omni-high-cpu-alarm

- **Metric:** CPU Utilization
- **Condition:** Greater than 80% for 1 datapoint
- **Period:** 5 minutes
- **Actions:** SNS notification

Alarm 2: omni-instance-health-alarm

- **Metric:** StatusCheckFailed
- **Condition:** Greater than 0 for 1 datapoint
- **Period:** 5 minutes
- **Actions:** SNS notification

Screenshot References:

- AWS-28-CloudWatch-CPU-Alarm.png - CPU alarm at 80%
- AWS-29-CloudWatch-Alarms-List.png - Both alarms listed

4.6.3 CloudWatch Dashboard

Dashboard Name: omni-security-dashboard

Widgets:

- CPU Utilization metric graph
- NetworkIn metric graph
- NetworkOut metric graph
- Text widget with project summary

Text Widget Content:

OMNI SECURITY PROJECT

AWS Resources:

- EC2: omni-server
- ALB: omni-alb
- WAF: omni-web-acl
- GuardDuty: Enabled

Alarms configured:

- High CPU (>80%)
- Instance health

Screenshot Reference: AWS-30-CloudWatch-Dashboard.png - omni-security-dashboard

4.7 Azure Monitoring (Basic)

4.7.1 Activity Log

Events Captured:

- Start Virtual Machine (Succeeded)
- Deallocate Virtual Machine (Succeeded)
- Create or Update Virtual Machine Extension (Failed)
- Health Event Updated

Screenshot Reference: Azure-07-Activity-Log.png - Activity log showing VM operations

4.7.2 VM Metrics

Metrics Collected:

- Network In Total
- Network Out Total
- Disk Read Bytes
- Disk Write Bytes

Screenshot Reference: Azure-08-VM-Metrics.png - VM metrics dashboard

4.7.3 Azure Alerts

Aler Rule: omni-high-cpu-alert

- **Condition:** CPU percentage greater than 80%
- **Actions:** Email notification
- **Action Group:** omni-email-group

Screenshot References:

- Azure-09-Alerts.png - Alerts page configured
- Azure-10-Email-Confirmation.png - Email confirmation for action group

5 Phase 3: DevSecOps Pipeline & Container Security

5.1 Overview

Phase 3 implements a complete DevSecOps pipeline with automated security scanning, containerization, and CI/CD automation using AWS services.

5.2 GitHub Repository Setup

- **Repository URL:** <https://github.com/SiddhantRangrao/omni-secure-app>
- **Repository Name:** omni-secure-app
- **Branches:** main (default)
- **Visibility:** Public

Repository Files:

- app.js - Node.js application with Express and Helmet.js
- package.json - Dependencies configuration
- Dockerfile - Container definition
- buildspec.yml - AWS CodeBuild specifications
- .dockerignore - Container build exclusions
- README.md - Project documentation

Screenshot Reference: AWS-38-GitHub-Repo.png - GitHub repository omni-secure-app

5.3 Application Code

5.3.1 Node.js Application (app.js)

```
1 const express = require('express');
2 const helmet = require('helmet');
3
4 const app = express();
5 const port = process.env.PORT || 3000;
6
7 // Use Helmet for security headers
8 app.use(helmet());
9
10 // Simple route
11 app.get('/', (req, res) => {
12   res.send('
13     <html>
14       <head><title>Omni Secure App</title></head>
15       <body>
16         <h1>      Omni Secure Application</h1>
17         <p>DevSecOps Pipeline Demo</p>
18         <p>Version: 1.0.0</p>
```

```

19     </ body >
20   </ html >
21   ');
22 });
23
24 // Health check endpoint
25 app.get('/health', (req, res) => {
26   res.status(200).json({ status: 'healthy' });
27 });
28
29 // Start server
30 app.listen(port, () => {
31   console.log('App listening at http://localhost:${port}');
32 });

```

Listing 7: Omni Secure Application Code

Screenshot Reference: AWS-39-App-Code.js - Application code with Helmet.js

5.3.2 Package.json

```

1 {
2   "name": "omni-secure-app",
3   "version": "1.0.0",
4   "description": "Secure sample app for DevSecOps demo",
5   "main": "app.js",
6   "scripts": {
7     "start": "node app.js",
8     "test": "echo \"No tests yet\" && exit 0"
9   },
10  "dependencies": {
11    "express": "4.18.2",
12    "helmet": "7.0.0"
13  },
14  "devDependencies": {
15    "mocha": "10.2.0"
16  }
17 }

```

Listing 8: Package.json Dependencies

Screenshot Reference: AWS-40-Package-JSON.png - package.json with dependencies

5.3.3 Dockerfile

```

1 FROM node:18-alpine
2
3 WORKDIR /usr/src/app
4
5 COPY package*.json ./
6
7 RUN npm ci --only=production
8
9 COPY . .
10
11 RUN addgroup -g 1001 -S nodejs && \
12     adduser -S nodejs -u 1001 && \
13     chown -R nodejs:nodejs /usr/src/app

```

```

14
15 USER nodejs
16
17 EXPOSE 3000
18
19 HEALTHCHECK --interval=30s --timeout=3s --start-period=5s --retries=3 \
20 CMD node -e "require('http').get('http://localhost:3000/health', (r) => {r
    .statusCode === 200 ? process.exit(0) : process.exit(1)})"
21
22 CMD ["node", "app.js"]

```

Listing 9: Dockerfile

5.4 AWS CodeBuild Configuration

5.4.1 CodeBuild Project Setup

- **Project Name:** omni-app-build
- **Source Provider:** GitHub
- **Repository:** SiddhantRangrao/omni-secure-app
- **Environment:** Ubuntu, Standard 7.0 image
- **Service Role:** codebuild-omni-app-build-service-role
- **Artifacts:** S3 bucket (omni-codebuild-artifacts-siddhantrangra)

Screenshot Reference: AWS-42-CodeBuild-Project.png - CodeBuild project configured

5.4.2 Buildspec.yml - Complete Configuration

```

1 version: 0.2
2
3 phases:
4   install:
5     runtime-versions:
6       nodejs: 18
7     commands:
8       - echo "Installing dependencies ..."
9       - npm ci
10      - echo "Installing security scanners ..."
11      - npm install -g snyk
12      - curl -sL https://raw.githubusercontent.com/aquasecurity/trivy/main/
        contrib/install.sh | sh -s -- -b /usr/local/bin
13
14   pre_build:
15     commands:
16       - echo "Logging in to Amazon ECR ..."
17       - aws ecr get-login-password --region $AWS_DEFAULT_REGION | docker
        login --username AWS --password-stdin $AWS_ACCOUNT_ID.dkr.ecr.
        $AWS_DEFAULT_REGION.amazonaws.com
18
19       - echo "Running security scans ..."
20       - echo "SCA Scan (Snyk) ..."

```

```

21     - SNYK_TOKEN=$(aws secretsmanager get-secret-value --secret-id omni/
    snyk-token --query SecretString --output text | jq -r .SNYK_TOKEN)
22     - snyk auth $SNYK_TOKEN
23     - snyk test --severity-threshold=high || true
24
25 build:
26   commands:
27     - echo "Building Docker image..."
28     - docker build -t $ECR_REPOSITORY:latest .
29     - echo "Scanning image with Trivy..."
30     - trivy image --severity HIGH,CRITICAL --no-progress $ECR_REPOSITORY:
    latest
31
32 post_build:
33   commands:
34     - echo "Pushing image to ECR..."
35     - docker tag $ECR_REPOSITORY:latest $AWS_ACCOUNT_ID.dkr.ecr.
    $AWS_DEFAULT_REGION.amazonaws.com/$ECR_REPOSITORY:latest
36     - docker push $AWS_ACCOUNT_ID.dkr.ecr.$AWS_DEFAULT_REGION.amazonaws.
    com/$ECR_REPOSITORY:latest
37     - echo "Build completed on 'date'"
38
39 artifacts:
40   files:
41     - '**/*'
42   base-directory: '.'
43   discard-paths: no

```

Listing 10: buildspec.yml with Security Scanning

5.5 Amazon ECR (Elastic Container Registry)

5.5.1 ECR Repository Creation

- **Repository Name:** omni-secure-app
- **Repository URI:** 778754217204.dkr.ecr.us-east-1.amazonaws.com/omni-secure-app
- **Tag Immutability:** Mutable
- **Encryption:** AES-256
- **Scan on Push:** Enabled

Screenshot Reference: AWS-43-ECR-Repository.png - ECR repository created

5.5.2 Images Pushed

- **Total Images:** 2
- **Tag:** latest
- **Created:** March 4, 2026

Screenshot Reference: AWS-44-ECR-Images.png - Images pushed to ECR with latest tag

5.6 AWS CodePipeline

5.6.1 Pipeline Configuration

- **Pipeline Name:** SimpleDockerService1
- **Source Stage:** GitHub (via GitHub App)
- **Build Stage:** Docker Build Tag and Push (CodeBuild)
- **Deploy Stage:** Skipped (optional)

Execution Status:

- **Source Stage:** ✓ Succeeded (22 minutes ago)
- **Build Stage:** ✓ Succeeded (Just now)

Screenshot Reference: AWS-45-CodePipeline-Success.png - CodePipeline successful execution

5.7 Security Scanning Implementation

Scan Type	Tool	Purpose
SCA (Software Composition Analysis)	Snyk	Dependency vulnerability scanning
Container Vulnerability Scanning	Trivy	Image vulnerability detection
Registry Scanning	ECR	Automated scan on push
SAST (Static Analysis)	ESLint	Code quality and security

Table 3: Security Scanning Tools Implemented

6 Phase 4: Attack Simulation & SOAR Automation

6.1 Overview

Phase 4 simulates real-world attacks and implements automated remediation using AWS services. Note: SSH attack was skipped as per project requirements.

6.2 Attack 1: Public S3 Bucket Simulation

6.2.1 Attack Bucket Creation

- **Bucket Name:** omni-attack-test-siddhant-rangra-public
- **Region:** us-east-1
- **Block Public Access:** Disabled (intentionally)
- **Purpose:** Simulate misconfigured S3 bucket

Screenshot Reference: AWS-46-Attack-Bucket-Created.png - Test bucket created for attack

6.2.2 Test File Upload

```
1 echo "This is a test file for public bucket attack" > test.txt
```

Listing 11: Test File Creation

- **File Name:** test.txt
- **File Size:** 62.0 B
- **Upload Date:** March 5, 2026

Screenshot Reference: AWS-47-Test-File-Uploaded.png - Test file uploaded to bucket

6.2.3 Public Policy Application

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": "*",
7       "Action": "s3:GetObject",
8       "Resource": "arn:aws:s3:::omni - attack - test - siddhant - rangra - public /*"
9     }
10  ]
11 }
```

Listing 12: Public Read Policy

Screenshot Reference: AWS-48-Public-Policy-Applied.png - Public read policy applied

6.2.4 Public Access Verification

Command:

```
1 curl https://omni-attack-test-siddhant-rangra-public.s3.amazonaws.com/test.txt
```

Result: File accessible publicly

Screenshot Reference: AWS-49-File-Public-Access.png - File accessible publicly

6.2.5 GuardDuty Detection

Findings Generated:

Finding Type	Severity	Description
Policy:S3/Bucket Block		S3 Block Public Access
	No Public Access	
Policy:S3/Bucket Access	Anonymous use access	S3 Public Anonymous Access was granted for the bucket

Table 4: GuardDuty Findings from Attack

Screenshot References:

- AWS-50-GuardDuty-Finding.png - Block Public Access disabled finding and Anonymous access granted finding

6.3 Attack 2: Unauthorized API Calls (Replaces SSH Attack)

6.3.1 Test User Creation

- **Username:** test-user
- **Permissions:** Limited (no termination or deletion rights)
- **Console Access:** Disabled
- **ARN:** arn:aws:iam::778754217204:user/test-user

Screenshot Reference: AWS-52-IAM-Test-User.png - Test user created

6.3.2 Unauthorized Actions Attempted

Attempt 1: Terminate EC2 Instance

```
1 aws ec2 terminate-instances --instance-ids i-123456789 abcdef0
```

Response: "An error occurred (InvalidInstanceID) when calling the Terminate Instances operation"

Attempt 2: Delete S3 Bucket

```
1 aws s3api delete-bucket --bucket omni-attack-test-siddhant-rangra-public
```

Response: "An error occurred (BucketNotEmpty) when calling the DeleteBucket operation"

Screenshot Reference: AWS-53-Unauthorized-Attempt.png - Failed termination/delete attempt

6.3.3 CloudTrail Event Logging

Events Captured:

- DeleteBucket (Access Denied)
- TerminateInstances (Access Denied)
- DeleteSession (CloudShell)
- PutCredentials (CloudShell)

Screenshot Reference: AWS-59-CloudTrail-Logs.png - CloudTrail showing events

6.4 SOAR Automation - Self-Healing Infrastructure

6.4.1 Lambda IAM Role Creation

- **Role Name:** omi-lambda-s3-remediation
- **Policies Attached:**
 - AWSLambdaBasicExecutionRole
 - s3-remediation (inline policy)

Inline Policy: s3-remediation

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "s3:PutBucketPublicAccessBlock",
8         "s3:PutBucketPolicy",
9         "s3:DeleteBucketPolicy",
10        "s3:GetBucketPolicy",
11        "s3:GetBucketPublicAccessBlock"
12      ],
13      "Resource": "*"
14    }
15  ]
16 }
```

Listing 13: s3-remediation Policy

Screenshot Reference: AWS-55-Lambda-Role.png - IAM role for Lambda remediation.

6.4.2 Lambda Function Deployment

Function Name: omni-fix-public-s3

Runtime: Python 3.9

Handler: lambda_function.lambda_handler

```

1 import boto3
2 import json
3 import logging
4
5 logger = logging.getLogger()
6 logger.setLevel(logging.INFO)
7
8 s3 = boto3.client('s3')
9
10 def lambda_handler(event, context):
11     logger.info(f"Received event: {json.dumps(event)}")
12
13     try:
14         if 'detail' in event and 'requestParameters' in event['detail']:
15             bucket_name = event['detail']['requestParameters']['bucketName']
16         else:
17             bucket_name = event.get('bucket_name', None)
18
19         if not bucket_name:
20             raise ValueError("Could not determine bucket name")
21
22         logger.info(f"Fixing public access for bucket: {bucket_name}")
23
24         response = s3.put_public_access_block(
25             Bucket=bucket_name,
26             PublicAccessBlockConfiguration={
27                 'BlockPublicAcls': True,
28                 'IgnorePublicAcls': True,
29                 'BlockPublicPolicy': True,
30                 'RestrictPublicBuckets': True
31             }
32         )
33
34         try:
35             s3.delete_bucket_policy(Bucket=bucket_name)
36             logger.info(f"Deleted bucket policy for {bucket_name}")
37         except:
38             logger.info(f"No bucket policy to delete for {bucket_name}")
39
40         return {
41             'statusCode': 200,
42             'body': f'Successfully remediated public access for bucket: {
bucket_name}'
43         }
44
45     except Exception as e:
46         logger.error(f"Error: {str(e)}")
47         return {
48             'statusCode': 500,
49             'body': f'Error: {str(e)}'
50         }

```

Listing 14: Lambda Remediation Code

Screenshot Reference: AWS-56-Lambda-Function.png - Lambda function code

6.4.3 Lambda Testing

Test Event:

```

1 {
2   "bucket_name": "omni-auto-fix-test-siddhant"
3 }

```

Test Result: ✓ Successfully saved

Screenshot Reference: AWS-57-Lambda-Test.png - Lambda test event saved

6.4.4 EventBridge Rule Configuration

Rule Name: s3-to-lambda

Description: Trigger Lambda when S3 bucket becomes public

Status: Enabled

Event Pattern:

```

1 {
2   "source": ["aws.s3"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventSource": ["s3.amazonaws.com"],
6     "eventName": ["PutBucketPolicy"]
7   }
8 }

```

Target: omni-fix-public-s3 Lambda function

Screenshot Reference: AWS-58-EventBridge-Rule.png - EventBridge rule created

6.4.5 Automation Test - Self-Healing Demonstration

Test Bucket: omni-auto-fix-test-siddhant

Process Flow:

1. Public policy applied to test bucket
2. EventBridge detected PutBucketPolicy event
3. Lambda function triggered automatically
4. Lambda executed remediation code
5. Bucket secured automatically

6.4.6 Verification - Bucket After Automation

Block Public Access: ON

Bucket Policy: "No policy to display"

Status: ✓ Secured

Screenshot Reference: AWS-61-Bucket-Fixed-After-Automation.png - Bucket secured after automation

6.4.7 Lambda Execution Logs

```

1 2026-03-06T02:30:17.878Z - Fixing public access for bucket: omni-auto-fix-
  test-siddhant
2 2026-03-06T02:30:17.967Z - No bucket policy to delete for omni-auto-fix-test
  -siddhant
3 Duration: 110.00 ms
4 Memory Used: 85 MB

```

Listing 15: CloudWatch Logs Showing Remediation

Screenshot Reference: AWS-62-Lambda-Logs-Remediation.png - Lambda execution logs

6.4.8 EventBridge Metrics Verification

Metric	Value	Description
MatchedEvents	>0	S3 events detected
TriggeredRules	>0	Rule activated
Invocations	>0	Lambda triggered
SuccessfulInvocations	>0	Lambda executed successfully
End-to-end latency	70-142 ms	Fast response time

Table 5: EventBridge Metrics After Automation

Screenshot References:

- AWS-63-EventBridge-Metrics-1.png - EventBridge MatchedEvents graph
- AWS-64-EventBridge-Metrics-2.png - EventBridge invocation metrics

7 Phase 5: Compliance & Cost Optimization

7.1 Overview

Phase 5 implements compliance monitoring using AWS Config and ensures all resources are optimized for cost.

7.2 AWS Config Compliance Monitoring

7.2.1 CloudTrail S3 Bucket for Logging

- **Bucket Name:** aws-cloudtrail-logs-778754217204-082dde55
- **Purpose:** Store CloudTrail logs for all API events
- **Region:** us-east-1
- **Creation Date:** March 5, 2026

Screenshot Reference: AWS-60-CloudTrail-S3-Bucket.png - CloudTrail S3 bucket

7.2.2 AWS Config Rules Evaluated

Rule Name	Status	Description
s3-bucket-public-read-prohibited	Evaluated	Prevents public read access to S3 buckets
iam-user-mfa-enabled	Evaluated	Ensures users have MFA enabled
cloud-trail-enabled	Evaluated	Verifies CloudTrail is logging
vpc-default-security-group-closed	Evaluated	Default security groups should restrict traffic

Table 6: AWS Config Rules Implemented

7.2.3 Compliance Results

- **Non-compliant Rules:** 1
- **Non-compliant Resources:** 2 (Default security groups with inbound/outbound rules)
- **Compliant Resources:** 9

Non-compliant Resources:

- sg-02edb9f228e8a7c2a - Default security group
- sg-0d4cb9005dc147f7a - Default security group

Screenshot Reference: AWS-65-Config-NonCompliant-Rules.png - AWS Config showing non-compliant resources

7.3 Cost Optimization Summary

7.3.1 AWS Cost Optimization

Resource	Status	Monthly Cost (Running)	Monthly Cost (Stopped)
EC2 (t2.micro)	STOPPED	\$8.50	\$2.50 (storage only)
ALB	Optional	\$20.00	\$0 (if deleted)
WAF	Optional	\$5.00	\$0 (if deleted)
EBS Storage	Active	\$2.00	\$2.00
S3 Buckets	Minimal	\$0.05	\$0.05
CloudWatch	Minimal	\$1.00	\$1.00
Total		\$36.55	\$5.55

Table 7: AWS Cost Analysis

7.3.2 Azure Cost Optimization

Resource	Status	Monthly Cost (Running)	Monthly Cost (Stopped)
VM (Standard_B1s)	STOPPED (deallocated)	\$14.00	\$0 (compute)
OS Disk (30 GB)	Active	\$1.50	\$1.50
Public IP	Released	\$3.00	\$0
Total		\$18.50	\$1.50

Table 8: Azure Cost Analysis

7.3.3 Total Cost Savings

- **Total Monthly Cost (All Running):** \$55.05
- **Total Monthly Cost (Optimized):** \$7.05
- **Monthly Savings:** \$48.00
- **Actual Project Cost:** \$2.60 (with cleanup)

7.3.4 Cost Optimization Techniques Implemented

- EC2 instances stopped when not in use
- Azure VM deallocated (no compute charges)
- Public IPs released when not needed
- Used free tier resources where available (t2.micro, B1s)
- No NAT Gateways or VPC Endpoints

- Single Availability Zone deployment
- Resources deleted after testing
- S3 lifecycle policies for log management

8 Screenshot Index (75 Files)

8.1 Phase 1 - AWS (9 files)

#	Filename	Description
1	AWS-01-VPC-Dashboard.png	VPC dashboard showing omni-vpc
2	AWS-02-Subnet.png	Public subnet configuration
3	AWS-03-Route-Table.png	Route table with IGW route
4	AWS-04-Security-Group.png	Security group inbound rules
5	AWS-05-Key-Pair.png	Key pair omni-key created
6	AWS-06-EC2-Running.png	EC2 instance in RUNNING state
7	AWS-07-EC2-SSH-Nginx-Install.png	SSH session with nginx installation
8	AWS-08-Nginx-Welcome-Page.png	Browser showing nginx welcome page
9	AWS-09-EC2-Stopped.png	EC2 instance in STOPPED state

Table 9: Phase 1 - AWS Screenshots

8.2 Phase 1 - Azure (6 files)

#	Filename	Description
1	Azure-01-Resource-Group.png	Resource group with all 6 resources
2	Azure-02-VNet.png	Virtual network omni-vnet
3	Azure-03-NSG-Rules.png	NSG inbound rules (ONLY port 443)
4	Azure-04-VM-Running.png	VM in RUNNING state
5	Azure-05-Port-443-Test.png	Port 443 connection successful
6	Azure-06-VM-Stopped.png	VM in STOPPED (deallocated) state

Table 10: Phase 1 - Azure Screenshots

8.3 Phase 2 - AWS (28 files: AWS-10 through AWS-37)

#	Filename	Description
---	----------	-------------

10	AWS-10-IAM-Group.png	User group omni-admins
11	AWS-11-IAM-User-Operator.png	User omni-operator created
12	AWS-12-IAM-Operator-Credentials.csv	Credentials for omni-operator
13	AWS-13-IAM-Viewer-Credentials.csv	Credentials for omni-viewer
14	AWS-14-IAM-Users-List.png	Both IAM users listed
15	AWS-15-IAM-JSON-Policy.png	Custom JSON policy for omni-readonly
16	AWS-16-IAM-Omni-Readonly-Policy.png	omni-readonly policy created
17	AWS-17-IAM-Role.png	Role omni-ec2-s3-readonly created
18	AWS-18-ALB-Created.png	Application Load Balancer omni-alb
19	AWS-19-ALB-Test.png	ALB showing nginx welcome page
20	AWS-20-WAF-Rules.png	Web ACL with 19 rules
21	AWS-21-WAF-Test-403.png	403 Forbidden from SQL injection
22	AWS-22-GuardDuty-Dashboard.png	GuardDuty dashboard
23	AWS-23-GuardDuty-Runtime.png	Runtime Monitoring enabled
24	AWS-24-GuardDuty-Settings.png	GuardDuty settings
25	AWS-25-SNS-Topic.png	SNS topic omni-alerts created
26	AWS-26-SNS-Email-Confirmation.png	Email confirmation request
27	AWS-27-SNS-Subscription.png	Subscription confirmed
28	AWS-28-CloudWatch-CPU-Alarm.png	CPU alarm at 80%
29	AWS-29-CloudWatch-Alarms-List.png	Both alarms listed
30	AWS-30-CloudWatch-Dashboard.png	omni-security-dashboard
31	AWS-31-Secrets-Manager.png	Secret omni/app/secrets stored
32	AWS-32-Secrets-Values.png	Secret values displayed

33	AWS-33-Secrets-CLI.png	CLI retrieval of secrets
34	AWS-34-Secrets-Policy.png	omni-secret-readonly policy
35	AWS-35-Secrets-Policy-Attached.png	Policy attached to user
36	AWS-36-MFA-Enabled.png	MFA enabled on root account
37	AWS-37-IAM-Test-Denied.png	omni-viewer denied EC2 creation

Table 11: Phase 2 - AWS Screenshots

8.4 Phase 2 - Azure (4 files)

#	Filename	Description
7	Azure-07-Activity-Log.png	Activity log showing VM start/stop
8	Azure-08-VM-Metrics.png	VM metrics (Network In/Out, Disk)
9	Azure-09-Alerts.png	Alerts page configured
10	Azure-10-Email-Confirmation.png	Email confirmation for action group

Table 12: Phase 2 - Azure Screenshots

8.5 Phase 3 - AWS (8 files: AWS-38 through AWS-45)

#	Filename	Description
38	AWS-38-GitHub-Repo.png	GitHub repository omni-secure-app
39	AWS-39-App-Code.js	Application code with Helmet.js
40	AWS-40-Package-JSON.png	package.json with dependencies
41	AWS-41-Buildspec-YML.png	Buildspec with security scans
42	AWS-42-CodeBuild-Project.png	CodeBuild project omni-app-build
43	AWS-43-ECR-Repository.png	ECR repository omni-secure-app
44	AWS-44-ECR-Images.png	Images pushed to ECR
45	AWS-45-CodePipeline-Success.png	CodePipeline successful execution

Table 13: Phase 3 - AWS Screenshots

8.6 Phase 4 - AWS (14 files: AWS-46 through AWS-59)

#	Filename	Description
46	AWS-46-Attack-Bucket-Created.png	Test bucket created for attack
47	AWS-47-Test-File-Uploaded.png	Test file uploaded to bucket
48	AWS-48-Public-Policy-Applied.png	Public read policy applied
49	AWS-49-File-Public-Access.png	File accessible publicly
50	AWS-50-GuardDuty-Finding-1.png	Block Public Access disabled finding
51	AWS-51-GuardDuty-Finding-2.png	Anonymous access granted finding
52	AWS-52-IAM-Test-User.png	Test user with limited permissions
53	AWS-53-Unauthorized-Attempt.png	Failed termination/delete attempts
54	AWS-54-CloudWatch-Alarms.png	CloudWatch alarms state
55	AWS-55-Lambda-Role.png	IAM role for Lambda remediation
56	AWS-56-Lambda-Function.png	Lambda function code
57	AWS-57-Lambda-Test.png	Lambda test event saved
58	AWS-58-EventBridge-Rule.png	EventBridge rule created
59	AWS-59-CloudTrail-Logs.png	CloudTrail showing events

Table 14: Phase 4 - AWS Screenshots

8.7 Phase 5 - AWS (5 files: AWS-60 through AWS-64)

#	Filename	Description
60	AWS-60-CloudTrail-S3-Bucket.png	CloudTrail S3 bucket for logging
61	AWS-61-Bucket-Fixed-After-Automation.png	Bucket secured after automation

62	AWS-62-Lambda-Logs-Remediation.png	Lambda execution logs showing fix
63	AWS-63-EventBridge-Metrics-1.png	EventBridge MatchedEvents graph
64	AWS-64-EventBridge-Metrics-2.png	EventBridge invocation metrics

Table 15: Phase 5 - AWS Screenshots

8.8 Phase 6 - AWS (1 file: AWS-65)

#	Filename	Description
65	AWS-65-Config-NonCompliant-Rules.png	AWS Config showing non-compliant resources

Table 16: Phase 6 - AWS Screenshots

9 Project Summary & Conclusions

9.1 Project Achievements

- **Total Screenshots:** 75 (65 AWS + 10 Azure)
- **Phases Completed:** 5 full phases + Phase 6 (Compliance)
- **AWS Services Used:** 20+ including VPC, EC2, IAM, WAF, Secrets Manager, GuardDuty, CloudWatch, Lambda, EventBridge, S3, CodeBuild, CodePipeline, ECR, AWS Config
- **Azure Services Used:** 5+ including Resource Group, VNet, NSG, VM, Activity Log, Metrics, Alerts
- **Security Controls:** 15+ including IAM least privilege, WAF, Secrets rotation, MFA, GuardDuty, CloudWatch alarms
- **Attack Simulations:** 2 (Public S3 + Unauthorized API calls)
- **SOAR Automations:** 1 (Self-healing S3 bucket remediation)
- **Compliance Rules:** 4 AWS Config rules evaluated

9.2 Key Technical Learnings

1. **Multi-Cloud Infrastructure:** Successfully deployed and managed resources across AWS and Azure, understanding each provider's unique approach to networking, security, and identity management.
2. **Identity & Access Management:** Implemented least privilege principles with custom IAM policies, groups, roles, and MFA enforcement. Demonstrated access denial for unauthorized actions.
3. **Web Application Firewall:** Configured WAF with managed rules protecting against SQL injection, XSS, and common web attacks. Verified protection through attack simulation.
4. **Secrets Management:** Implemented secure credential storage with AWS Secrets Manager, including CLI-based retrieval and access policies.
5. **Threat Detection:** Enabled GuardDuty with runtime monitoring, successfully detecting S3 misconfigurations and generating security findings.
6. **Monitoring & Alerting:** Created comprehensive CloudWatch dashboards, SNS notifications, and Azure monitoring for complete visibility across both clouds.
7. **DevSecOps Pipeline:** Built automated CI/CD pipeline with integrated security scanning using Snyk (SCA) and Trivy (container scanning), ensuring vulnerabilities are caught early.
8. **Container Security:** Implemented Docker best practices including non-root users, health checks, and ECR vulnerability scanning.

9. **SOAR Automation:** Designed and implemented self-healing infrastructure using EventBridge and Lambda, reducing incident response time from hours to minutes.
10. **Compliance Monitoring:** Established continuous compliance monitoring using AWS Config, identifying non-compliant resources automatically.
11. **Cost Optimization:** Implemented resource stopping/deallocation strategies, achieving over 85% cost reduction compared to running all resources continuously.

9.3 Challenges Faced & Solutions

Challenge	Impact	Solution
Azure IAM Limitations	Could not use Azure AD for identity federation	Used AWS as primary identity hub, Azure for resource-only deployment
SSH Attack Simulation	SSH port was properly secured (good security)	Replaced with unauthorized API call attack (CloudTrail logging)
EventBridge Rule Configuration	Initial rule didn't trigger Lambda	Simplified event pattern and enabled CloudTrail logging
Lambda Permissions	Function couldn't modify S3 buckets	Added inline policy with necessary S3 permissions
Cost Management	Resources could run continuously	Implemented stop/start strategy and automated cleanup

Table 17: Challenges and Solutions

9.4 Best Practices Demonstrated

1. **Least Privilege Access:** IAM policies granting only necessary permissions
2. **Defense in Depth:** Multiple security layers (WAF, IAM, GuardDuty, NSG)
3. **Shift-Left Security:** Security scanning integrated into CI/CD pipeline
4. **Immutable Infrastructure:** Containerized applications with versioned images
5. **Infrastructure as Code:** Documented all steps for future automation
6. **Continuous Monitoring:** CloudWatch, GuardDuty, and Azure Metrics
7. **Automated Remediation:** SOAR workflows for common security issues
8. **Compliance Automation:** AWS Config for continuous compliance checking
9. **Cost Governance:** Resource tagging and stop/start strategies
10. **Comprehensive Documentation:** 75 screenshots documenting every step

9.5 Future Enhancements

- **Terraform Conversion:** Convert manual steps to Infrastructure as Code
- **Expanded SOAR:** Add automated responses for more attack types
- **Multi-Region Deployment:** Deploy across multiple AWS regions
- **Advanced Compliance:** Map controls to NIST/CIS frameworks
- **Threat Intelligence:** Integrate with threat intelligence feeds
- **Kubernetes Integration:** Deploy application to EKS/AKS
- **Service Mesh:** Implement Istio for mTLS and traffic policies
- **Chaos Engineering:** Test resilience with controlled failures

9.6 Final Project Statistics

Metric	Value
Total Screenshots	75
AWS Screenshots	65
Azure Screenshots	10
Total Phases	5 (plus Phase 6)
AWS Services Used	20+
Azure Services Used	5+
Security Controls Implemented	15+
Attack Simulations	2
SOAR Automations	1
Compliance Rules	4
Total Project Cost	\$2.60
Monthly Cost (Optimized)	\$7.05
Monthly Savings vs Running	\$48.00
Documentation Pages	20+

Table 18: Final Project Statistics

9.7 Conclusion

The Omni-Cloud Security Project successfully demonstrated a comprehensive, multi-cloud security implementation across AWS and Azure. Through five structured phases, the project established:

- **Secure Infrastructure:** Isolated VPC/VNet environments with proper network controls
- **Identity Security:** IAM best practices including least privilege and MFA
- **Application Security:** WAF protection against common web attacks

- **Secrets Management:** Secure credential storage and retrieval
- **Threat Detection:** Continuous monitoring with GuardDuty and CloudWatch
- **DevSecOps:** Automated CI/CD pipeline with integrated security scanning
- **Attack Simulation:** Real-world attack scenarios with successful detection
- **SOAR Automation:** Self-healing infrastructure reducing response time
- **Compliance:** Automated compliance monitoring with AWS Config
- **Cost Optimization:** Effective resource management saving over 85%

With 75 screenshots documenting every step, this project serves as a comprehensive reference for multi-cloud security implementation. The total cost of \$2.60 demonstrates that comprehensive security testing and learning can be achieved economically with proper resource management.

9.8 References

- AWS Documentation: <https://docs.aws.amazon.com>
- Azure Documentation: <https://docs.microsoft.com/en-us/azure>
- GitHub Repository: <https://github.com/SiddhantRangra0/omni-secure-app>
- Snyk Documentation: <https://docs.snyk.io>
- Trivy Documentation: <https://aquasecurity.github.io/trivy>

A Appendix A: AWS CLI Commands Used

VPC Commands

```
aws ec2 create-vpc --cidr-block 10.0.0.0/16 --tag-specifications 'ResourceType=vpc,Tags=
aws ec2 create-subnet --vpc-id vpc-xxx --cidr-block 10.0.1.0/24 --availability-zone us-e
aws ec2 create-internet-gateway
aws ec2 attach-internet-gateway --vpc-id vpc-xxx --internet-gateway-id igw-xxx
aws ec2 create-route-table --vpc-id vpc-xxx
aws ec2 create-route --route-table-id rtb-xxx --destination-cidr-block 0.0.0.0/0 --gatew
```

EC2 Commands

```
aws ec2 create-key-pair --key-name omni-key --query 'KeyMaterial' --output text > omni-k
aws ec2 run-instances --image-id ami-xxx --instance-type t2.micro --key-name omni-key
```

IAM Commands

```
aws iam create-group --group-name omni-admins
aws iam attach-group-policy --group-name omni-admins --policy-arn arn:aws:iam::aws:polic
aws iam create-user --user-name omni-operator
aws iam add-user-to-group --user-name omni-operator --group-name omni-admins
aws iam create-policy --policy-name omni-readonly --policy-document file://policy.json
```

```
# Secrets Manager
aws secretsmanager create-secret --name omni/app/secrets --secret-string file:///secrets.
aws secretsmanager get-secret-value --secret-id omni/app/secrets --query 'SecretString'

# S3 Commands
aws s3 mb s3://omni-attack-test-siddhant-rangra-public
aws s3api put-bucket-policy --bucket omni-attack-test-siddhant-rangra-public --policy
aws s3 cp test.txt s3://omni-attack-test-siddhant-rangra-public/

# Lambda Commands
aws lambda create-function --function-name omni-fix-public-s3 --runtime python3.9 --role
```

B Appendix B: Azure CLI Commands Used

```
# Resource Group
az group create --name omni-project-rg --location eastus2

# VNet and Subnet
az network vnet create --resource-group omni-project-rg --name omni-vnet --address-prefi

# NSG and Rules
az network nsg create --resource-group omni-project-rg --name omni-nsg
az network nsg rule create --resource-group omni-project-rg --nsg-name omni-nsg --name

# VM Creation
az vm create --resource-group omni-project-rg --name omni-vm --image Ubuntu2204 --size

# VM Stop/Deallocate
az vm deallocate --resource-group omni-project-rg --name omni-vm

# Monitor Metrics
az monitor metrics list --resource /subscriptions/xxx/resourceGroups/omni-project-rg/pro
```

C Appendix C: GitHub Repository Structure

```
omni-secure-app/
  app.js
  package.json
  Dockerfile
  buildspect.yml
  .dockerignore
  README.md
```